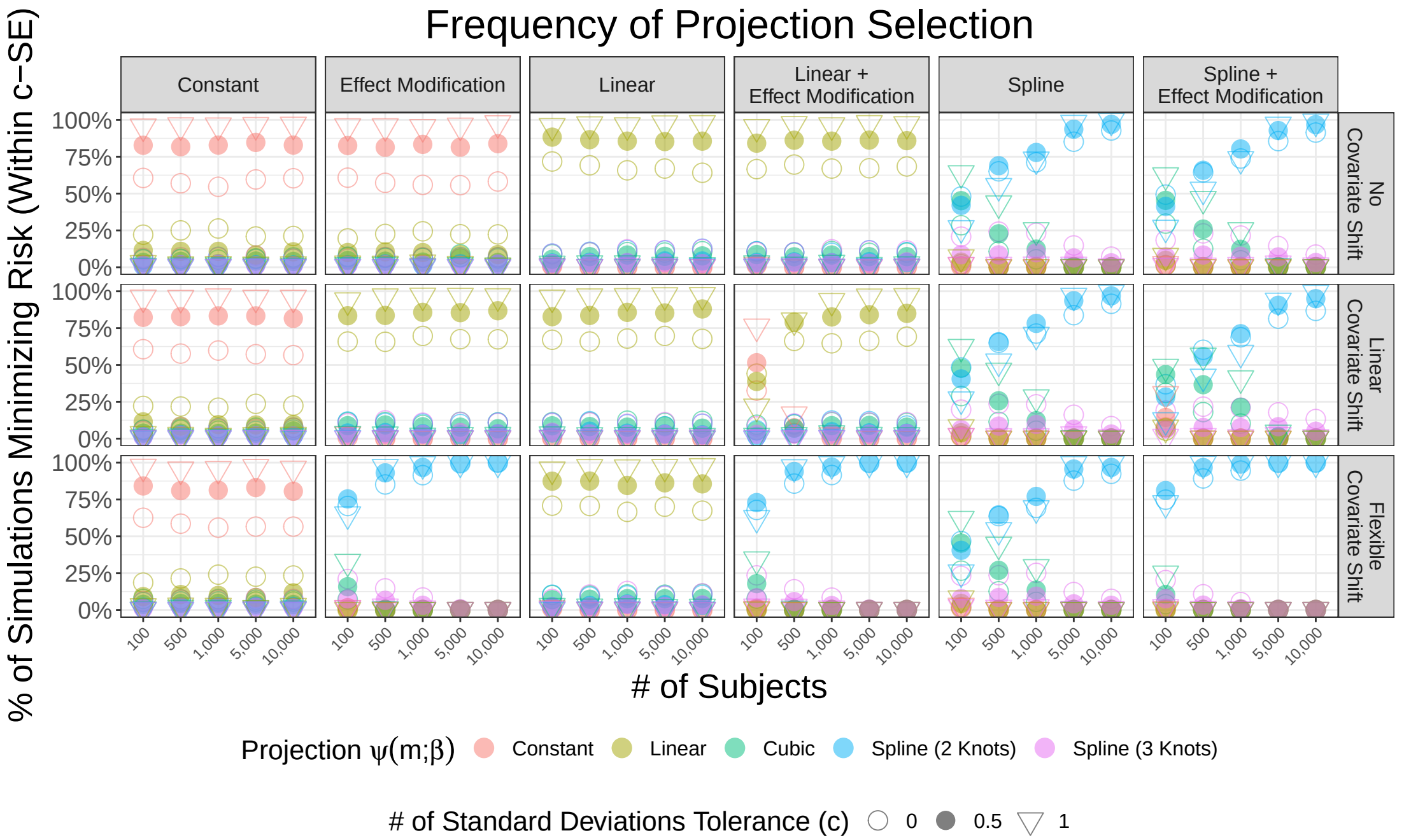
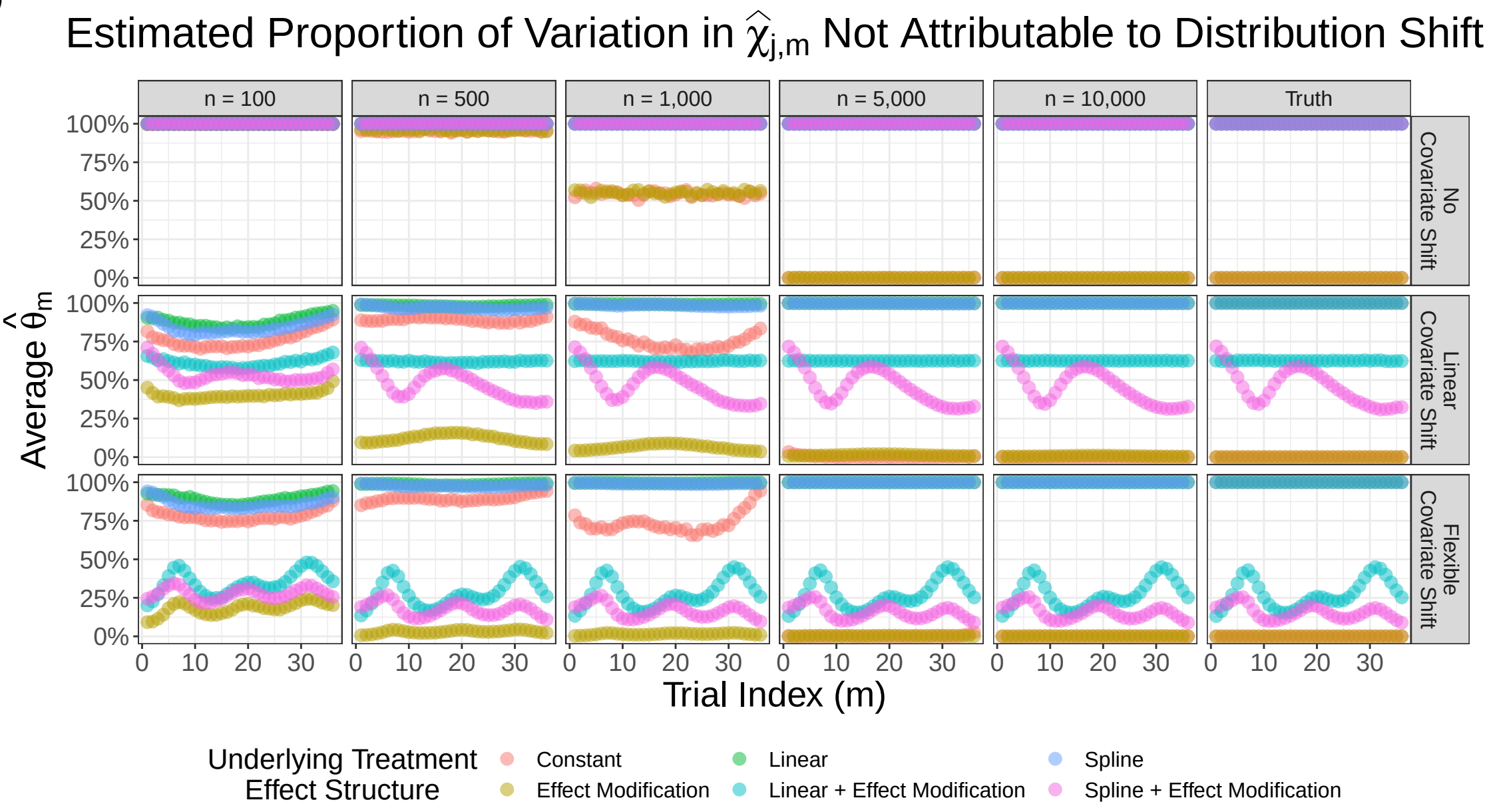
A)


Frequency of Projection Selection
Constant
Effect Modification
Linear
Linear + Effect Modification
Spline
Spline + Effect Modification
No Covariate Shift
Linear Covariate Shift
Flexible Covariate Shift
100%
75%
50%
25%
0%
100
500
1,000
5,000
10,000
% of Simulations Minimizing Risk (Within c−SE)
of Subjects
Projection ψ(m;β)
Constant
Linear
Cubic
Spline (2 Knots)
Spline (3 Knots)
of Standard Deviations Tolerance (c)
0
0.5
1

B)


Estimated Proportion of Variation in $\hat{\chi}_{j,m}$ Not Attributable to Distribution Shift
n = 100
n = 500
n = 1,000
n = 5,000
n = 10,000
Truth
No Covariate Shift
Linear Covariate Shift
Flexible Covariate Shift
100%
75%
50%
25%
0%
Average $\hat{\theta}_m$
0
10
20
30
Trial Index (m)
Underlying Treatment Effect Structure
Constant
Effect Modification
Linear
Linear + Effect Modification
Spline
Spline + Effect Modification